

Experiment-4B

Aim: To develop a login system that verifies user email and password securely using a database.

Description:

Project Setup

```
npm init -y
```

```
npm install express mongoose body-parser bcryptjs
```

Folder Structure

```
login-app/
```

```
|— server.js
```

```
|— models/User.js
```

```
|— createUser.js
```

Program:

Server.js

```
const express = require("express");
const mongoose = require("mongoose");
const bodyParser = require("body-parser");
const bcrypt = require("bcryptjs");
const User = require("../models/User.js");
const app = express();
app.use(bodyParser.urlencoded({ extended: true }));
app.use(bodyParser.json());
// MongoDB connection
mongoose.connect("mongodb://127.0.0.1:27017/loginDB")
  .then(() => console.log("MongoDB Connected"))
  .catch(err => console.log(err));
// Login route
app.post("/login", async (req, res) => {
  const { email, password } = req.body;

  try {
    const user = await User.findOne({ email });
    if (!user) {
      return res.status(400).send("User not found");
    }
  }
});
```

```

    }
    const isMatch = await bcrypt.compare(password, user.password);
    if (!isMatch) {
        return res.status(400).send("Invalid password");
    }
    res.send("Login successful");
} catch (err) {
    res.status(500).send("Server error");
}
});
// Server start
app.listen(3000, () => {
    console.log("Server running on http://localhost:3000");
});

```

Models/User.js

```

const mongoose = require("mongoose");
const UserSchema = new mongoose.Schema({
    email: {
        type: String,
        required: true,
        unique: true
    },
    password: {
        type: String,
        required: true
    }
});
module.exports = mongoose.model("User", UserSchema);

```

CreateUser.js

```

const mongoose = require("mongoose");
const bcrypt = require("bcryptjs");
mongoose.connect("mongodb://127.0.0.1:27017/loginDB")
    .then(() => console.log("MongoDB Connected"));
async function createUser() {
    const passwordHash = await bcrypt.hash("12345", 10);

    const db = mongoose.connection;

```

```
await db.collection("users").insertOne({
  email: "admin@gmail.com",
  password: passwordHash
});
console.log("User created successfully");
process.exit();
}
createUser();
```

Output:

Testing Login (Postman / Form)

POST <http://localhost:3000/login>

